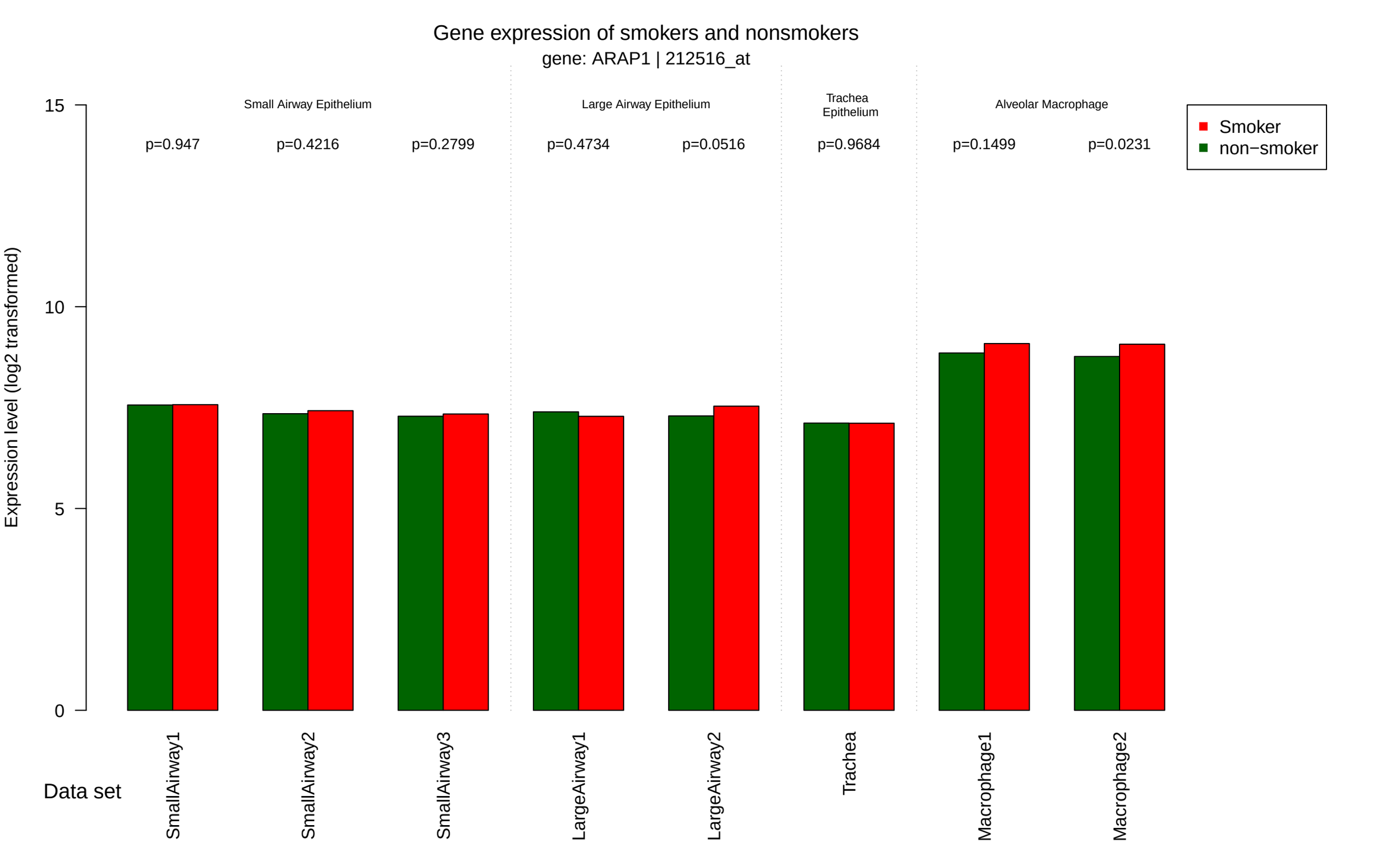
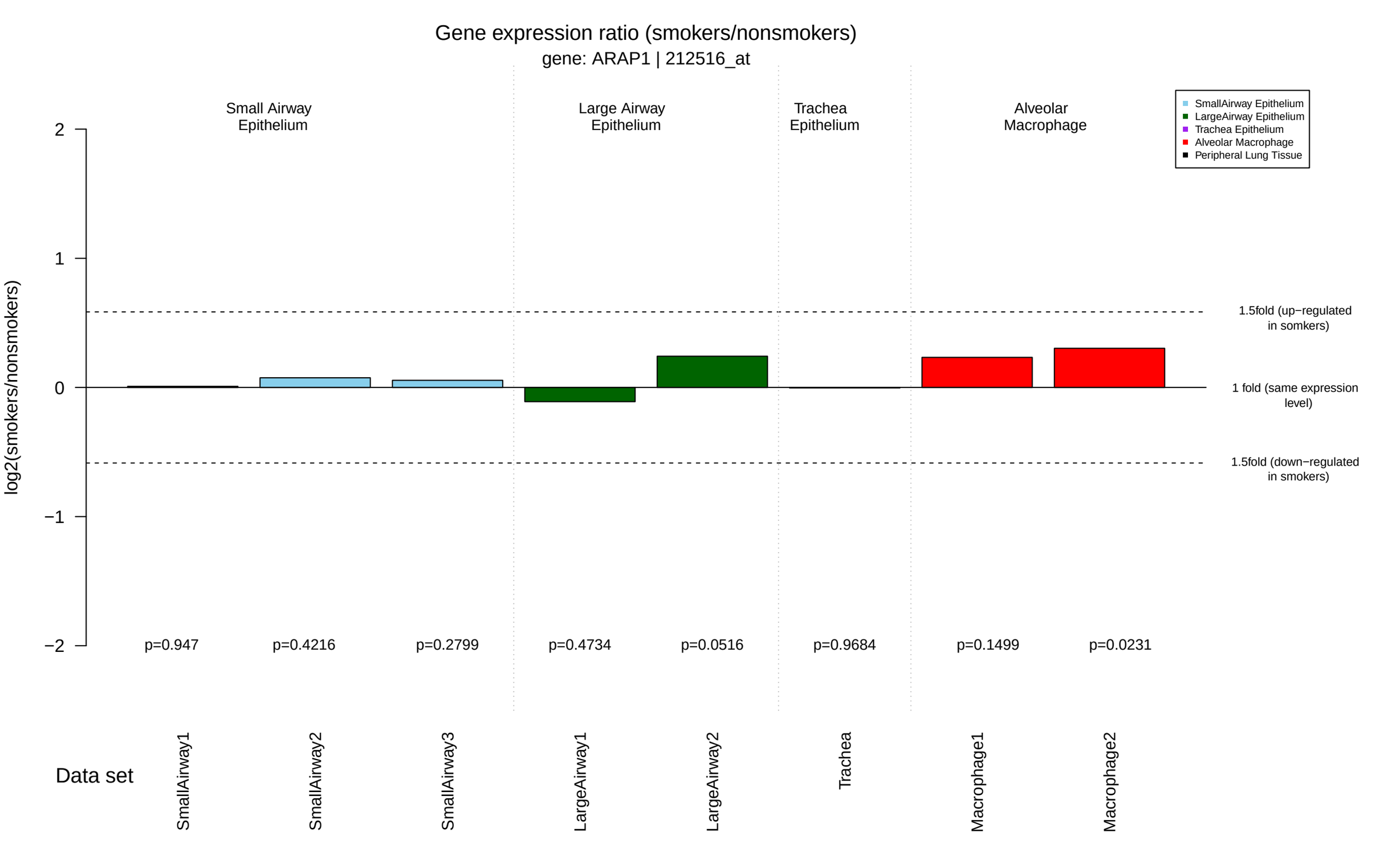
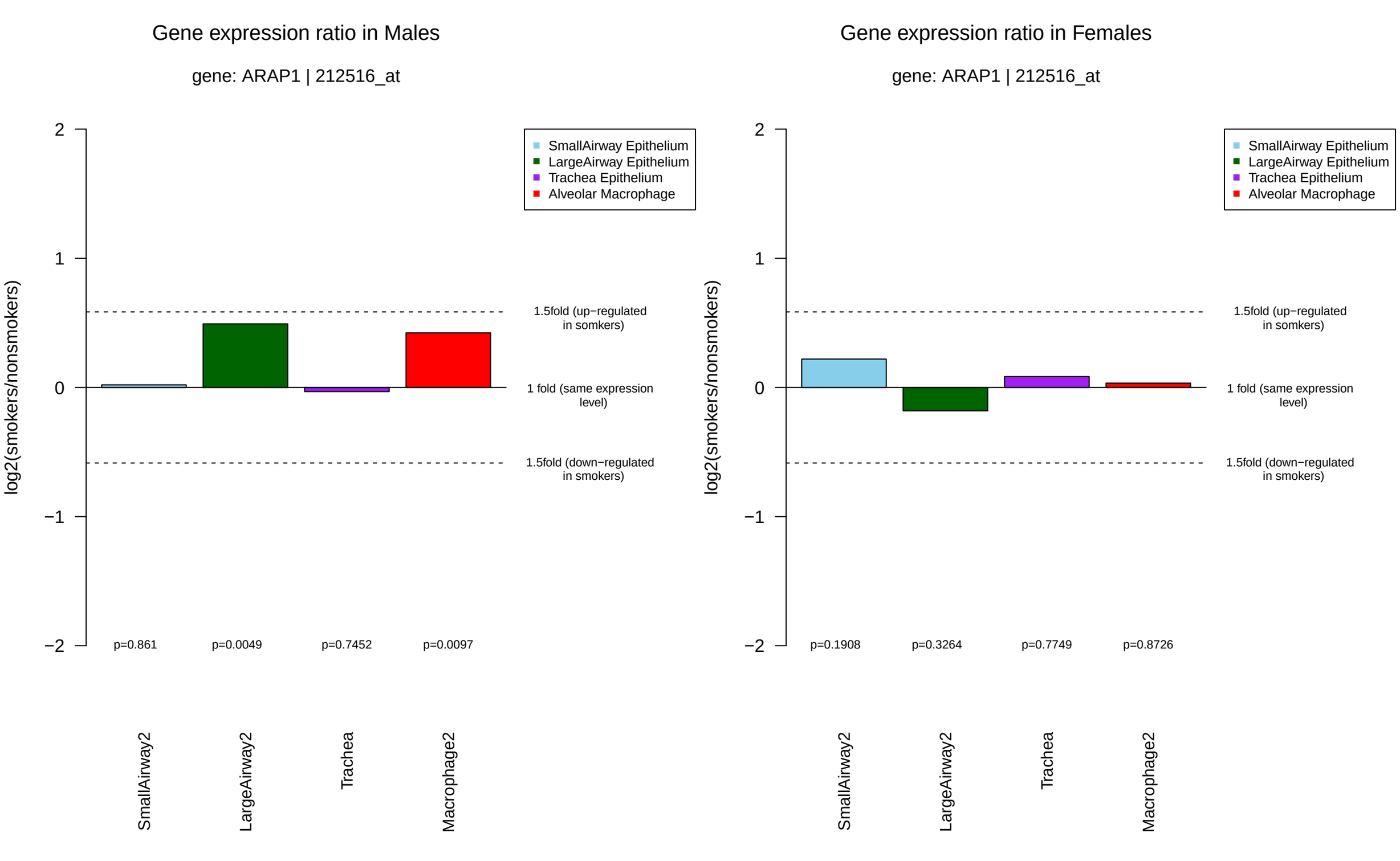
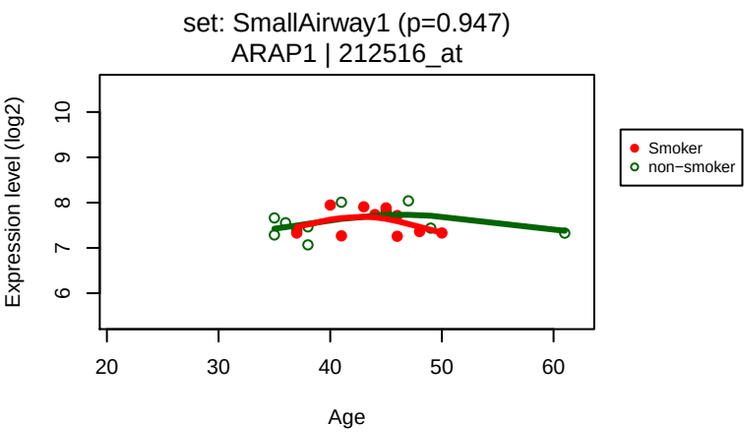
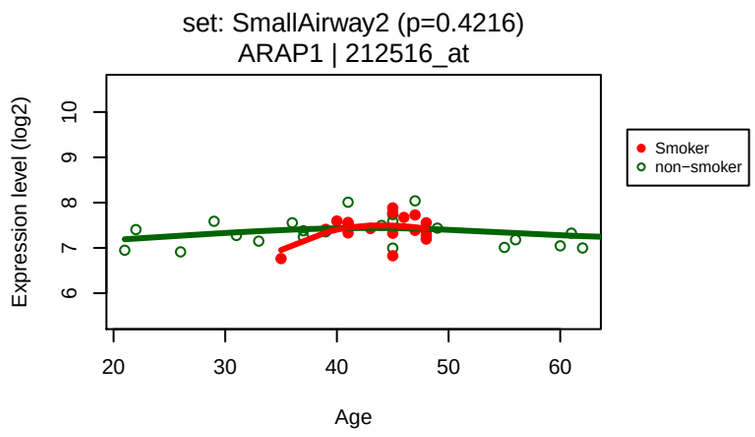
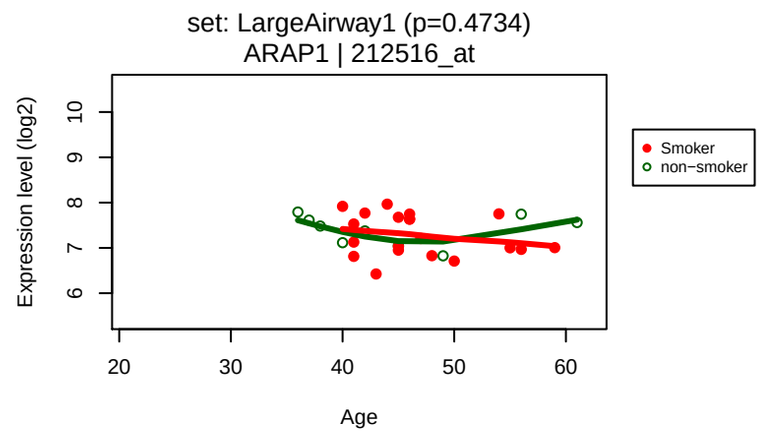

Gene expression of smokers and nonsmokers
gene: ARAP1 | 212516_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.947
p=0.4216
p=0.2799
p=0.4734
p=0.0516
p=0.9684
p=0.1499
p=0.0231
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

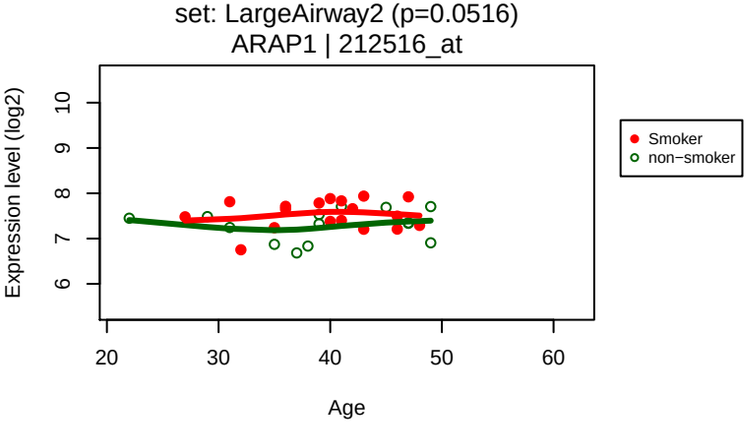

Gene expression ratio (smokers/nonsmokers)
gene: ARAP1 | 212516_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.947
p=0.4216
p=0.2799
p=0.4734
p=0.0516
p=0.9684
p=0.1499
p=0.0231
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

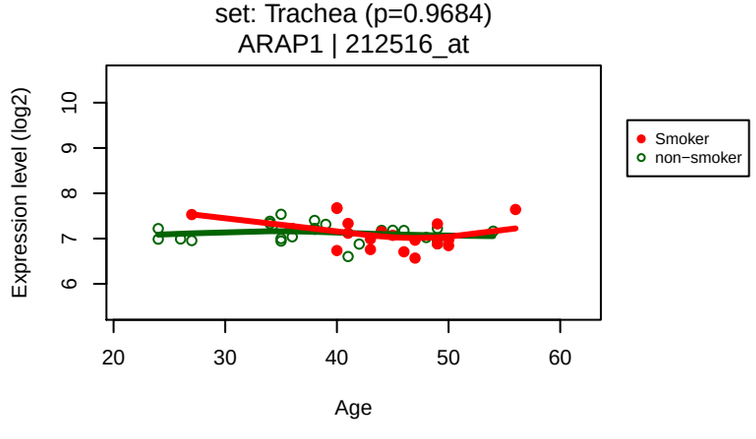

Gene expression ratio in Males
gene: ARAP1 | 212516_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.861
p=0.0049
p=0.7452
p=0.0097
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARAP1 | 212516_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1908
p=0.3264
p=0.7749
p=0.8726
SmallAirway2
LargeAirway2
Trachea
Macrophage2

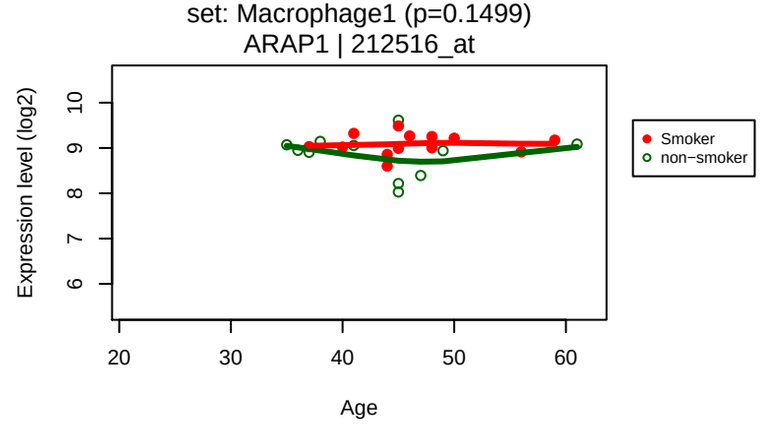

set: SmallAirway1 (p=0.947)
ARAP1 | 212516_at
Expression level (log2)
Age
Smoker
non−smoker

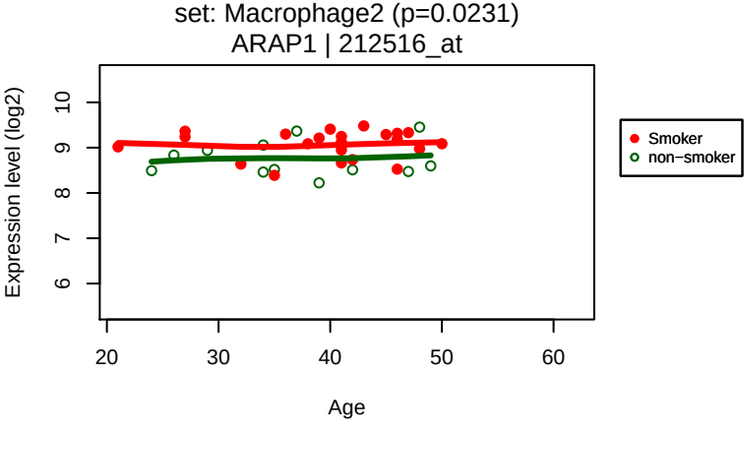

set: SmallAirway2 (p=0.4216)
ARAP1 | 212516_at
Expression level (log2)
Age
Smoker
non−smoker

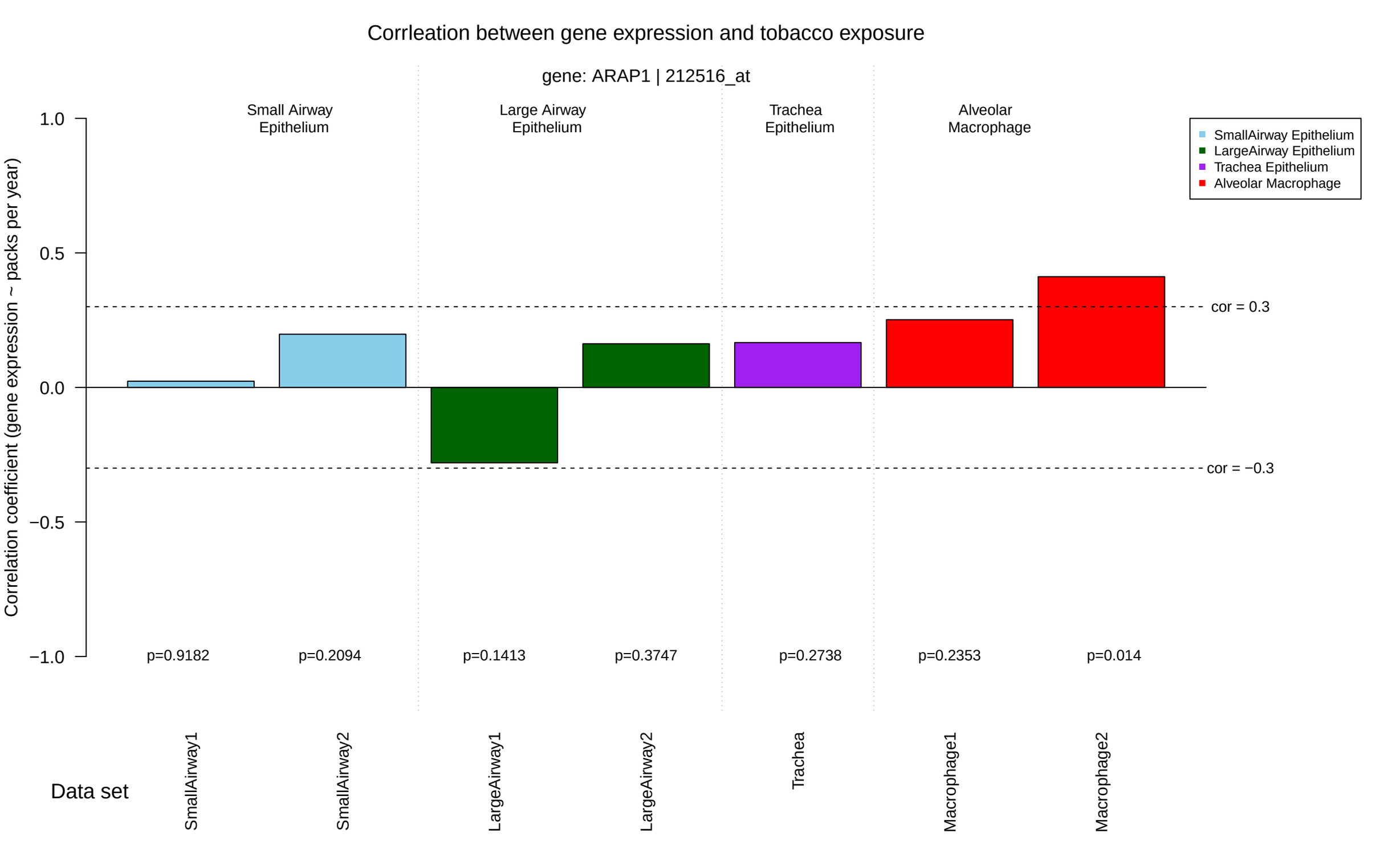

set: LargeAirway1 (p=0.4734)
ARAP1 | 212516_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0516)
ARAP1 | 212516_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9684)
ARAP1 | 212516_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1499)
ARAP1 | 212516_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0231)
ARAP1 | 212516_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARAP1 | 212516_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9182
p=0.2094
p=0.1413
p=0.3747
p=0.2738
p=0.2353
p=0.014
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2